

1. Smart Attendance Checker

```
import java.util.Scanner
public class AttendanceChecker
{
    public static void main (String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter total classes conducted:");
        int total = sc.nextInt();
        System.out.print("Enter classes attended:");
        int attended = sc.nextInt();
        double pct = ((double)attended/total)*100;
        System.out.println("Attendance = " + pct);

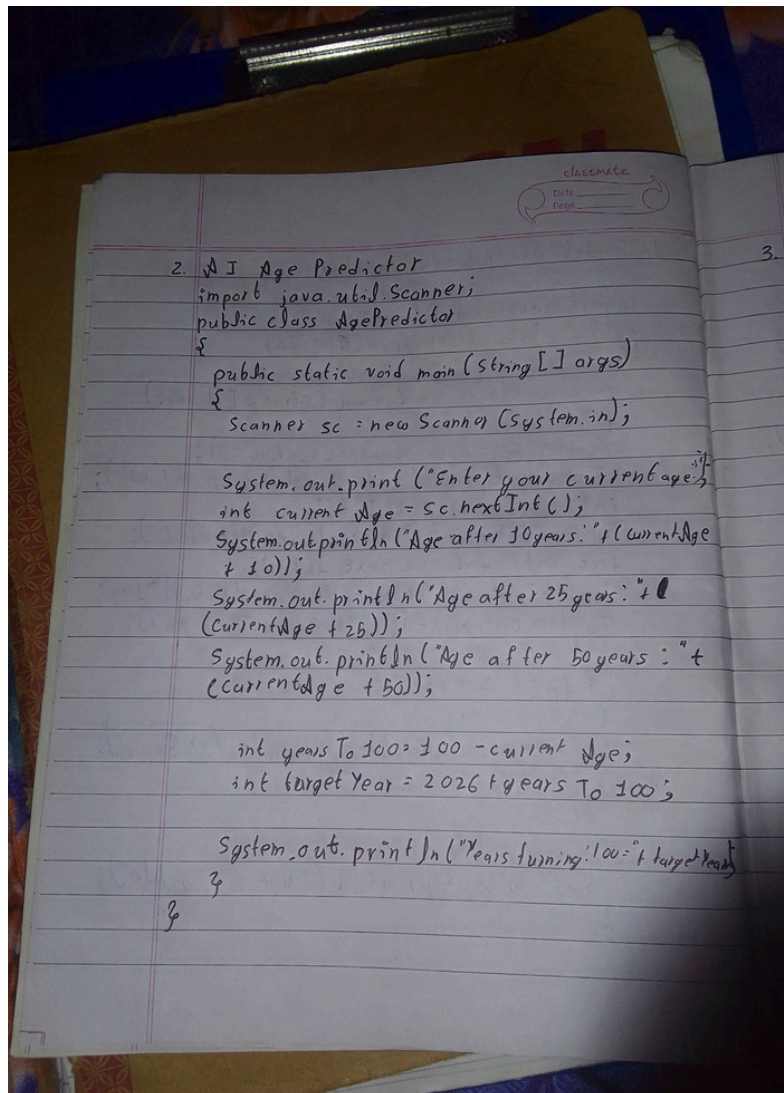
        if (pct >= 75)
        {
            System.out.println("Eligible for Exam");
        }
        else
        {
            System.out.println("Not Eligible");
        }
    }
}
```

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"
```

```
Enter marks obtained : 34
```

```
Your Grade is = F
```

```
Process finished with exit code 0
```



```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "  
Enter your current age : 19  
Age after 10 years : 29  
Age after 25 years : 44  
Age after 50 years : 69  
Year turning 100 = 2107  
  
Process finished with exit code 0
```


3. Fitness Challenge Tracker

```
import java.util.Scanner;
public class FitnessTracker
{
    public static void main (String[] args)
    {
        Scanner sc = new Scanner (System.in);
        int[] steps = new int[7];
        int total = 0;

        for (int i = 0; i < 7; i++)
        {
            System.out.print("Enter steps for day " + (i+1) + ": ");
            steps[i] = sc.nextInt();
            total = total + steps[i];
        }

        double avg = (double) total / 7;
        int max = steps[0];

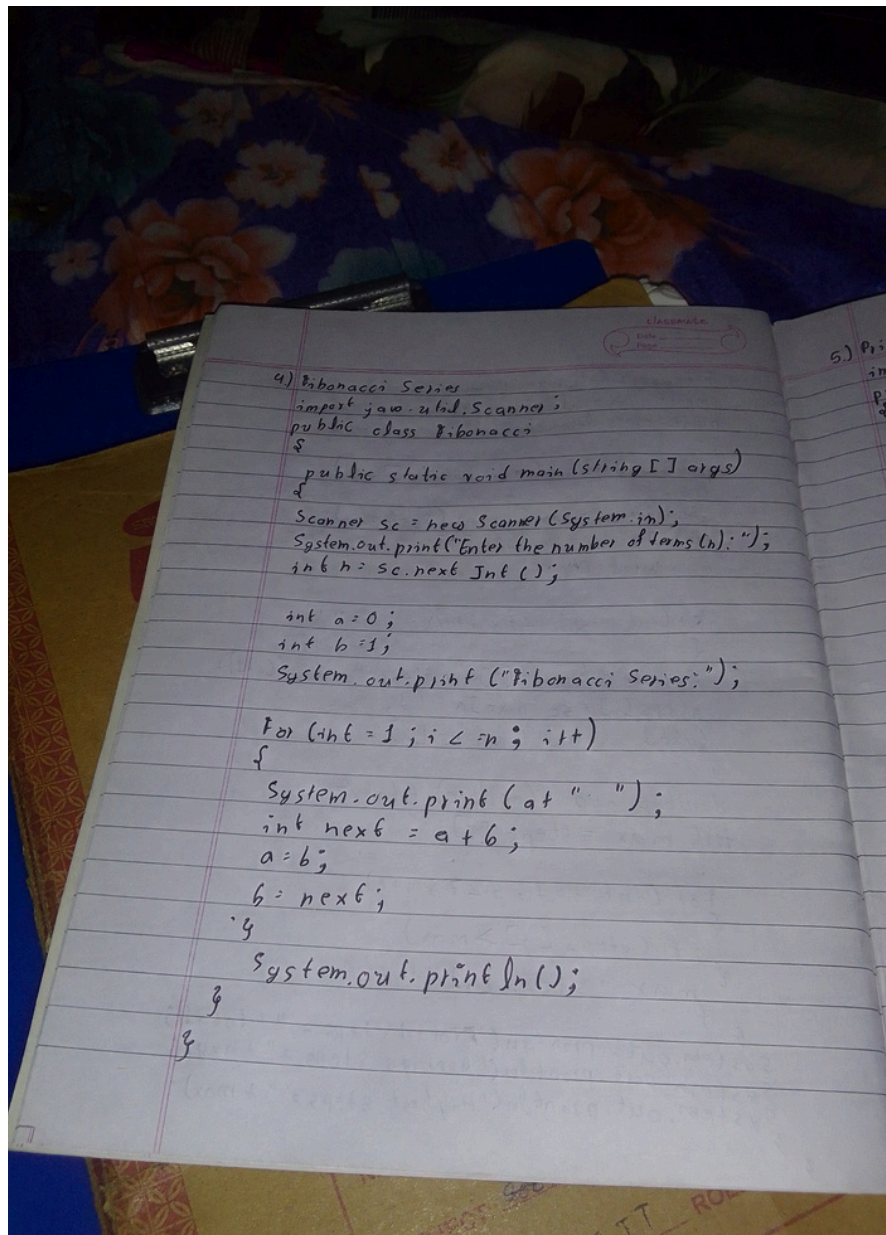
        for (int i = 1; i < 7; i++)
        {
            if (steps[i] > max)
            {
                max = steps[i];
            }
        }

        System.out.println("Total steps = " + total);
        System.out.println("Average steps = " + avg);
        System.out.println("Highest steps = " + max);
    }
}
```

SUBJECT: SOC

CLASS: BCS.IT

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"  
Enter steps for day 1 : 5677  
Enter steps for day 2 : 8999  
Enter steps for day 3 : 6666  
Enter steps for day 4 : 888  
Enter steps for day 5 : 9897  
Enter steps for day 6 : 89898  
Enter steps for day 7 : 99  
Total steps = 122124  
Average steps = 17446.285714285714  
Highest steps = 89898  
  
Process finished with exit code 0
```



```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"  
Enter the number of terms (n) : 2  
Fibonacci Series : 0 1  
  
Process finished with exit code 0
```


5.) Prime Number Check

```

import java.util.Scanner;
public class PrimeCheck
{
    public static void main (String[] args)
    {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter a number to check : ");
        int num = sc.nextInt();
        boolean isPrime = true;
        if (num <= 1)
        {
            isPrime = false;
        }
        else
        {
            for (int i = 2; i <= num/2; i++)
            {
                if (num % i == 0)
                {
                    isPrime = false;
                    break;
                }
            }
        }
        if (isPrime)
        {
            System.out.println (num + " is a prime number.");
        }
        else
        {
            System.out.println (num + " is not a prime number.");
        }
    }
}

```

```

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"
Enter a number to check : 89
89 is a prime number.

Process finished with exit code 0

```

```

6. Logical operators Check
import java.util.Scanner;
public class LogicalCheck
{
    public static void main (String [] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter First number :");
        int n1 = sc.nextInt();
        System.out.print("Enter Second number :");
        int n2 = sc.nextInt();
        System.out.print("Enter third number :");
        int n3 = sc.nextInt();

        boolean check = (n1 + n2 == n3);

        System.out.println("Third number is sum of
        first two = " + (check & & true));
    }
}

```

```

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "
Enter first number : 5
Enter second number : 6
Enter third number : 11
Third number is sum of first two = true

Process finished with exit code 0
|

```


7. Multiplication Table

```
import java.util.Scanner;
public class multitable
{
    public static void main (String [] args)
    {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter a number :");
        int num = sc.nextInt();

        System.out.println ("Multiplication Table of"
            + "\n" + num + " :");
        for (int i = 1; i <= 10; i++)
        {
            System.out.println (num + "x" + i + " = " + (num*i));
        }
    }
}
```



```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "
```

```
Enter a number : 1
```

```
Multiplication Table of 1 :
```

```
1 x 1 = 1
```

```
1 x 2 = 2
```

```
1 x 3 = 3
```

```
1 x 4 = 4
```

```
1 x 5 = 5
```

```
1 x 6 = 6
```

```
1 x 7 = 7
```

```
1 x 8 = 8
```

```
1 x 9 = 9
```

```
1 x 10 = 10
```

```
Process finished with exit code 0
```

classmate
Date _____
Page _____

8.) Right-Angled Triangle Pattern

```
import java.util.Scanner;  
public class TrianglePattern  
{  
    public static void main (String[] args)  
    {  
        Scanner sc = new Scanner (System.in);  
        System.out.print("Enter number of rows:");  
        int rows = sc.nextInt();  
  
        For (int i = 1; i <= rows; i++)  
        {  
            For (int j = 1; j <= i; j++)  
            {  
                System.out.print("\n");  
            }  
        }  
    }  
}
```



```
C:\Program Files\Java\jdk-20.0.1\bin\java.exe
Enter number of rows : 3
*
**
***

Process finished with exit code 0
|
```

9 Pyramid Pattern

```
import java.util.Scanner;
public class Pyramid Pattern
{
    public static void main (String [] args)
    {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter number of rows:");
        int rows = sc.nextInt();
```

```
        for (int i=1; i <= rows; i++)
        {
            for (int j=1; j <= rows - i; j++)
            {
                System.out.print (" ");
            }
            for (int k=1; k <= (2*i - 1); k++)
            {
                for (int k=1; k <= (2*i - 1); k++)
                {
                    System.out.print ("*");
                }
                System.out.println();
            }
        }
    }
}
```

```
}
```

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"
```

```
Enter number of rows : 6
```

```
    *
```

```
   ***
```

```
  *****
```

```
 *****
```

```
*****
```

```
*****
```

```
Process finished with exit code 0
```


10) Palindrome Number

```
import java.util.Scanner;
public class PalindromeCheck
{
    public static void main (String [] args)
    {
        Scanner sc = new Scanner (System.in);
        System.out.print("Enter an integer: ");
        int x = sc.nextInt();

        System.out.println("Is palindrome = " + isPalindrome(x));
    }

    public static boolean isPalindrome (int x)
    {
        if (x < 0) return false;
        int original = x;
        int reversed = 0;
        while (x != 0)
        {
            int digit = x % 10;
            reversed = reversed * 10 + digit;
            x /= 10;
        }
        return original == reversed;
    }
}
```

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"
```

```
Enter an integer : 5
```

```
Is palindrome = true
```

```
Process finished with exit code 0
```

```
|
```


11. Basic Calculator

```
import java.util.Scanner;
public class Basic Calculator
{
    public static void main (String[] args)
    {
        Scanner sc = new Scanner (System.in);

        System.out.println("Enter first number:");
        double num1 = sc.nextDouble();
        System.out.println("Enter an operator (+, -, *, /):");
        char operator = sc.next().charAt(0);
        System.out.println("Enter second number:");
        double num2 = sc.nextDouble();

        if (operator == '+') System.out.println("Result = " + (num1 + num2));
        if (operator == '-') System.out.println("Result = " + (num1 - num2));
        if (operator == '*') System.out.println("Result = " + (num1 * num2));
        if (operator == '/')
        {
            if (num2 != 0)
            {
                System.out.println("Result = " + (num1 / num2));
            }
            else
            {
                System.out.println("Error: Division by zero");
            }
        }
    }
}
```



```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "  
Enter first number : 3  
Enter an operator (+, -, *, /) : -  
Enter second number : 1  
Result = 2.0  
  
Process finished with exit code 0  
|
```

```

12. Grade Calculator
import java.util.Scanner;
public class GradeCalculator
{
    public static void main (String[] args)
    {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter marks obtained : ");
        double marks = sc.nextDouble();

        String grade = (marks >= 80) ? "A" :
            (marks >= 75) ? "A-" :
            (marks >= 70) ? "B+" :
            (marks >= 65) ? "B" :
            (marks >= 60) ? "C+" :
            (marks >= 55) ? "C" :
            (marks >= 50) ? "C-" :
            (marks >= 45) ? "D+" :
            (marks >= 40) ? "D" : "F";

        System.out.println ("Your Grade is : " + grade);
    }
}

```

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe"
```

```
Enter marks obtained : 68
```

```
Your Grade is = B
```

```
Process finished with exit code 0
```

```
|
```